MILITARY SPECIFICATION

PUMPING ASSEMBLY, LIQUID BULK TRANSFER, DIESEL-ENGINE-DRIVEN, 350 GPM CAPACITY, 275 TOTAL HEAD, WHEEL MOUNTED

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a wheel-mounted, direct-connected, diesel-engine-driven centrifugal pumping assembly.

1.2 Classification. Pumping assemblies shall be of the following types and styles as specified (see 6.2).

Type I - Pumping Assembly, POL

Style A - Without regulator (TA13220E1070).
Style B - With regulator (TA13226E2289).

Type II - Pumping Assembly, Water (TA13225E9200).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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PPP-P-40 - Packaging and Packing of Hand Tools.

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MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-G-1086 - Gasket Material, Synthetic Rubber (for Bolted Steel Tanks).
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-G-3859 - Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-C-6183 - Cork and Rubber Composition Sheet; for Aromatic Fuel and Oil Resistant Gaskets.
MIL-S-7916 - Sealing Compound, Thread and Gasket, Fuel, Oil and Water Resistant.
MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-G-20141 - Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-L-21260 - Lubrication Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-27487 - Coupling Halves, Quick-Disconnect, Cam-Locking Type.
MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-E-52649 - Engine Cold Starting Aids, Ether Fuel Primers.
MIL-V-58039 - Valves, Gate, Rising Stem, Double Acting, Aluminum.
MIL-S-81733 - Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-171	- Finishing of Metal and Wood Surfaces.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1400	- Engines, Gasoline or Diesel, Method of Test.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13220E1070	- Pumping Assembly, POL, Wheel Mounted, 4 Inch, 350 GPM at 275 Ft. Head.
TA13226E2289	- Pumping Assembly, POL, Wheel Mounted, 4 Inch, 350 GPM at 275 Ft. Head with Regulator.
TA13225E9200	- Pumping Assembly, Water Wheel Mounted, 4 Inch, 350 GPM at 275 Ft. Head.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

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HYDRAULIC INSTITUTE (HI)

Rating Standards and Test Code Standards.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, Cleveland, OH 44107).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096).

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pumping assembly shall consist of a diesel-engine-driven, centrifugal, self-priming pump with two 4-inch discharge ports and two 4-inch suction ports and shall be wheel mounted with all necessary accessories and connections in accordance with Top Assembly TA13220E1070, TA13226E2289, or TA13225E9200 and as specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of pumping assemblies from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish pumping assemblies conforming to this specification.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Material deterioration and control. The pumping assembly shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the pumping assembly may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pumping assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the pumping assembly produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the pumping assembly.

3.4 Gaskets. All cork and rubber composition gaskets shall conform to MIL-C-6183, type optional, class I, grade B, except that the gasket between the intermediate and pump volute, and the gasket on which the discharge check valve seats, shall be synthetic rubber conforming to MIL-G-1086, type I. Synthetic rubber gasket material shall meet all requirements of MIL-G-1086 except that the low temperature tests shall be conducted at -25° F and the hardness shall be 65-75 Shore A.

3.5 Threaded fasteners. All threaded fasteners shall be steel and zinc coated in accordance with MIL-STD-171, finish 1.9.2.3 or cadmium coated in accordance with MIL-STD-171, finish 1.1.2.3. Threads shall conform to FED-STD-H28.

3.6 Couplings. The quick-disconnect couplings shall conform to MIL-C-27487, type, size, style, and class as specified on applicable drawings.

3.7 Valves. The gate valves shall conform to MIL-V-58039, size 4-inch, type I. The gate valves shall have labels to indicate open-close and the direction to turn labels must conform to MIL-STD-1472, 5.4.1.2.4.

3.8 Environmental.

3.8.1 Operating temperature. The pumping assembly shall perform as specified herein in any ambient temperature from +120° F to -25° F.

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3.8.2 Storage temperature. The pumping assembly shall not be damaged by storage in any ambient temperature from +155° F to -65° F.

3.9 Safety. All shafts or rotating parts of the pumping unit shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Adequate shielding may be used to attain the same purpose. Exhausts or discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.

3.10 Human factors engineering. The pumping assembly shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, 4 (general requirements), 5.1 (control/display integration), 5.2 (visual displays), 5.4 (controls), 5.5 (labeling), 5.6 (anthropometry), 5.9 (design for maintainability), 5.13 (hazards and safety), of MIL-STD-1472, as applicable. An operation information plate conforming to drawing 13226E0111 shall be permanently placed near the control panel so that it can be seen while operating the pump.

3.10.1 Noise limits. The noise produced by the pumping assembly shall conform with MIL-STD-1474 5.1 requirements when tested in accordance with 4.5.2.6.1. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign. Plates shall be in accordance with MIL-P-514, type III, composition C of type I, grade A, class 1 material.

3.11 Reliability. The specified mean-time-between-failure (MTBF) shall be 200 hours when the pumping assembly is operated at rated capacity and tested as specified in 4.6.2.7.

3.12 Maintainability.

3.12.1 Maintenance support. Assemblages or support elements such as technical manuals, repair pares, special tools, or lubrication charts, as specified (see 6.5 and 6.6), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.8.

3.12.2 Maintenance ratio. The pumping assembly shall have a maintenance ratio of not more than 0.07. Maintenance ratio is defined as a ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

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3.13 Performance. The pumping assemblies, type I and type II, shall prime within 2 minutes, and deliver either liquid hydrocarbon fuels, having a specific gravity of 0.75, or water, having a specific gravity of 1.0 at not less than 350 gallons per minute (gpm) against a total dynamic head of 275 feet at 2400 revolutions per minute (rpm), +50 rpm. The maximum net positive suction head (NPSH) requirements shall not exceed 8 feet at 350 gpm at 275 feet total head at the operating speed of the pumping assembly. The pumping assembly shall develop a peak efficiency of not less than 63 percent.

3.13.1 Performance data chart. A performance data chart approximately 8.5 x 11 inches shall be furnished. The chart shall show the complete performance characteristics of the pumping assembly operation when operating at 2400 rpm and pumping liquid petroleum fuel having a specific gravity of 0.75 or water having a specific gravity of 1.0. All data shall be derived from the actual performance of the first article model. The curves will be plotted as total head, feet, US capacity, US gallons per minute. The additional scales will be presented to show the total head as psi of 1.0 and 0.75 specific gravity fluid. The chart shall be a photosensitive anodized aluminum plate conforming to MIL-P-514. The chart shall be located on the outside of the toolbox.

3.14 Pump. The pump shall be centrifugal and self-priming. The pump housing, check valve discharge housing, discharge housing, discharge pipe, and manifold inlet shall be high-strength aluminum alloy. Aluminum castings shall be uniform in quality and condition, free from cracks, hard spots, porosity, shrinkage defects, misplaced coring, and other defects which may affect the intended use.

3.14.1 Hydrostatic pressure. The pumping assembly, with the 4-inch valves completely open and with the quick-disconnect fitting covers in place, shall withstand an internal hydrostatic pressure of 200 pounds per square inch (psi) for not less than 5 minutes without rupture or distortion to any component, and without any leakage through the pump, fittings, piping, or joints.

3.14.2 Pump housing. The pump housing shall be provided with a provision for filling the pump housing without removal of the suction or discharge connections, and a drain port and plug capable of passing 0.25 inch solids at its lowest point for removing sediment and for draining the case without tilting.

3.14.3 Pump impeller. The impeller shall be as specified on the drawings.

3.14.3.1 Impeller balance. The impeller shall be dynamically balanced to a maximum out-of-balance condition of 0.25 inch-ounce at 2400 rpm.

3.14.4 Pump shaft seal. The pump shaft seal shall be a self-lubricating mechanical seal that will permit operation of the pumping assembly at any operating condition specified herein.

3.14.5 Valves (suction and discharge). The pumping assembly shall be equipped with four 4-inch valves as specified herein. The valves shall be secured to the frame of the unit or auxiliary members secured thereto. The discharge of the pump shall be manifolded into two 4-inch valves. Suction couplers and discharge

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adapters shall be slanted downward 30 degrees from the horizontal. Two 4-inch valves shall control the flow of liquid through the suction inlet. One of these 4-inch valves shall be in line with the inlet of the pump. All valves shall be clearly labeled to indicate their function (i.e., intake, discharge, etc.) and open-close directions. Direction for rotation to open and close shall be indicated with a double-ended arrow.

3.14.5.1 Check valve, suction. A suction check valve shall not be used on the inlet side of the pumping assembly.

3.14.6 Manifold. The manifold system shall insure interchangeability of all 4-inch valves. Each flange on the pump and manifold shall be equipped with a gasket 0.062 inch thick.

3.14.7 Fittings. Prior to assembly, threads on all screwed fittings shall be coated with sealing compound conforming to MIL-S-7916.

3.15 Diesel engine. For the purpose of selecting a diesel engine, the pumping assembly shall be classified as a class II end item in accordance with MIL-STD-1410. The engine shall be furnished with a 12-volt electric cranking and battery charging system; cold starting aid in accordance with MIL-E-52649, type III, electrically operated; and controls in accordance with MIL-STD-1410.

3.15.1 Engine operational requirements. The engine operational requirements and maximum engine operating limits shall meet the requirements and conditions specified herein.

3.15.2 Fuel tank. A fuel tank shall be furnished conforming to the drawings and as specified herein. The tank shall be equipped with a cap and strainer. The tank shall be provided with a hand-operated fuel shutoff valve attached directly to the tank and a mechanically actuated fuel gage mounted on the tank. The fuel tank outlet shall be not less than 0.50 inch above the bottom of the tank. At the tank's fuel outlet and fuel return openings or in their respective lines, furnish 3-way selector valves with the required adapter fittings. A corrosion-resisting circular metal face plate shall be mounted on the fuel selector valve and shall be marked "OFF", "UNIT TANK", and "AUXILIARY". The fuel tank shall not leak when tested as specified in 4.5.2.13. The words "DIESEL OR JET FUEL ONLY" shall be conspicuously marked or stenciled on or near the fuel cap in 0.75-inch letters. The auxiliary outlet shall have a cap to prevent dirt from entering the fuel tank. These caps shall be captive in order to prevent loss. The tank shall have a drain plug which is located to completely drain the tank.

3.16 Instruments. The instruments as specified on the drawings shall be securely mounted on a panel secured to the pumping assembly. Instruments shall be of moistureproof and dustproof construction with plastic lenses. Gauges shall be marked in English and metric units. All instruments and controls shall be clearly and permanently marked as to their function. The instrument panel shall be shock mounted to effectively withstand both road shock and operating vibration. The

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panel shall be located in a position so as to be readable when operating the suction and discharge valves and shall be positioned to afford the maximum protection to the instrument panel. The throttle shall be positioned to be operable when reading the instruments. The back of the instrument panel shall inclosed or fitted with a guard to protect the operator from possible shock or shorting of electrical connections and to protect instruments and controls fr damage. Instrument panel shall be illuminated as specified in 3.10. The pum assembly is a protected asset in accordance with visual displays of MIL-STD-14

3.17 Battery boxes. The battery boxes shall have a label which pictorially indicates battery connections.

3.18 Mobility. The pumping assembly shall be capable of being towed over r as follows without malfunction, permanent deformation, or damage to the pumpin assembly or its components:

a. Over hard-surfaced roads at an average speed of 20 miles per hour (mph
b. Over gravel roads at an average speed of 10 mph.
c. Over rough cross country terrain at an average speed of 8 mph.

3.19 Slope capability. The pumping assembly shall be capable of negotiatin side slopes up to 20 percent and approach, negotiate, and depart on a ramp of to 30 percent slope without contact of any component (except tires) with the r

3.20 Fording capability. Components of the pumping assembly that could be damaged by water immersion shall be either waterproofed or located not less th 30 inches above ground level to allow fording in water to that depth without w damage to components or leakage of any seals located at any point less than 30 inches above ground level.

3.21 Electromagnetic interference. The electromagnetic interference emissi characteristics of the pumping assembly shall conform to MIL-STD-461, class II

3.22 Lubrication. The pumping unit shall operate as specified herein when serviced with the following military lubricants, grades as applicable to the operating conditions:

Lubricating oil:	MIL-L-2104 or MIL-L-46167
Grease:	MIL-G-10924 or MIL-G-23827

The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated before delivery and shall be tagged conspicuous place to indicate the temperature range and grade of lubricant use

3.22.1 Lubrication fittings. The lubrication fittings shall be accessible hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility t lubrication fittings,

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3.22.2 Pressure-release device. A pressure release device shall be provided where use of pressure lubricating equipment will damage grease seals.

3.23 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.23.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus-and moisture-resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.23.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition IX material complies with Rule 442, South Coast Air Quality Management District.

3.24 Slinging provisions. Permanently affixed slinging provisions that enable the pumping assembly to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209.

3.25 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the pumping assembly to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209.

3.26 Transportability.

3.26.1 Rail. Unless otherwise specified (see 6.2), the unpackaged pumping assembly shall withstand rail humping without damage or permanent deformation to any part or component when tested as specified in 4.5.2.20. After testing, the pumping assembly shall operate as specified herein for a period of not less than 1 hour without malfunction or permanent deformation of any part of the pumping assembly.

3.27 Treatment and painting. The portions of the pumping assembly normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 conforming to MIL-C-46168.

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3.28 Workmanship.

3.28.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges, and corners (see 3.10), except when sharp edges or corners are required, or where they are not detrimental to safety.

3.28.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to witness or to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.8 and 6.9.1).

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4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article (reproduction) inspection.

4.3.1 Examination. The first article pumping assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection, or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first article pumping assembly shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection or shall be cause for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Tests.

4.4.3.1 Individual. Each pumping assembly and component shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table I. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The pumping assembly shall be examined as specified herein for the following defects:

Major

101. Any part or component not in accordance with the QAP requirements as shown on the drawings.
102. Material not as specified.
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.

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104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes, or treatment.
106. Used, rebuilt or remanufactured components, pieces and parts incorporated in the pumping assembly.
107. Components missing or not as specified. Each component missing or not as specified shall be considered one defect.
108. Parts or components do not function as specified.
109. Gaskets not as specified.
110. Threaded fasteners not as specified.
111. Couplings not as specified.
112. Valves not as specified.
113. Human factors not as specified.
114. Pump housing not as specified.
115. Pump impeller not as specified.
116. Pump shaft seal not as specified.
117. Instruments not as specified.
118. Battery boxes not as specified.
119. Electromagnetic interference suppression components and network not identical to the approved model.
120. Lubricants not as specified.
121. Lubrication fittings not as specified.
122. Pressure-release device not as specified.
123. Fungus and moisture resistance not as specified.
124. Workmanship not as specified.
125. Metal fabrication not as specified.
126. Bolted connections not as specified.
127. Any dimensions, other than those identified on the QAP requirements shown on the drawings, not as specified.
128. Data charts not as specified or missing.
129. Treatment and painting not as specified.
130. Paint color not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with the applicable provisions of the test code for centrifugal pumps of the standards of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions of barometric pressure of 29.92 inches of mercury and liquid fuel at 60° F having a specific gravity of 0.75 and vapor pressure of 4.0 psi. Water at a temperature of between 60° F and 80° F may be used as the test fluid.

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4.5.2.2 Test schedule.

TABLE 1. Test schedule.

First article examination	Quality Conformance: Individual	Quality Conformance: Sample	Tests	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
			Preassembly:		
X	X	-	Pump	4.5.2.9.1	3.13
X	X	-	Impeller	4.5.2.11	3.14.3.1
X	-	X	Engine	4.5.2.12	3.15
X	X	-	Fuel tank	4.5.2.13	3.15.2
X	-	-	Environmental	4.5.2.3	3.8
X	-	-	Safety	4.5.2.4	3.9
X	-	-	Human factors	4.5.2.5	3.10
X	-	-	Noise level measurement	4.5.2.6	3.10.1
X	-	-	Reliability	4.5.2.7	3.11
X	-	-	Maintainability	4.5.2.8	3.12
			Performance:		
X	-	X	Pumping assembly	4.5.2.9.2	3.13, 3.15.1
X	-	X	Priming	4.5.2.9.3	3.13
X	X	-	Hydrostatic pressure	4.5.2.10	3.14.1
X	-	-	Mobility	4.5.2.14	3.18
X	-	-	Slope	4.5.2.15	3.19
X	-	-	Fording	4.5.2.16	3.20
X	-	-	Electromagnetic interference	4.5.2.17	3.21
X	-	-	Slinging provisions	4.5.2.18	3.24
X	-	-	Tiedown provisions	4.5.2.19	3.25
X	-	-	Rail impact	4.5.2.20	3.26.1

4.5.2.3 Environmental.

4.5.2.3.1 High temperature. The pumping assembly shall be tested as specified in MIL-STD-810, procedure I, method 501. The maximum ambient storage temperature shall be 155° F and the maximum ambient operating temperature shall be 120° F. The operating period of procedure I shall be 2 hours. The pumping assembly shall be operated at rated capacity. Nonconformance to 3.8 shall constitute failure of this test.

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4.5.2.3.2 Low temperature. The pumping assembly shall be tested as specified in MIL-STD-810, procedure I, method 502. The minimum ambient storage temperature shall be -65° F. The minimum ambient operating temperature shall be -25° F. The operating period of procedure I shall be 2 hours. The pumping assembly shall be operated at rated capacity. Nonconformance to 3.8 shall constitute failure of this test.

4.5.2.4 Safety. The pumping assembly shall be evaluated for safety requirements throughout testing as specified in 4.5.2. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.5 Human factors. The human factors requirements shall be evaluated during testing. Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.6 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load, MIL-STD-1474, 5.1.2.1.4; contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position; additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.7 Reliability. Using the MTBF specified in 3.11, the pumping assembly shall be tested as specified in 4.5.2.9.2 with “accept” and “reject” criteria in accordance with figure 1. Continue testing until either an “accept” or “reject” decision is reached. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustment, repair, or replacement action using controls and on-equipment tools or parts available to the using element, and which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the pumping assembly by continued operation.
c. Safety hazards to personnel.

Nonconformance to 3.11 shall constitute failure of this test.

4.5.2.8 Maintainability.

4.5.2.8.1 Maintenance support evaluation. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.12.1. Failure of the support elements to maintain the pumping assembly as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.5.2.8.2 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Nonconformance to 3.12.2 shall constitute failure of this test.

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4.5.2.9 Performance.

4.5.2.9.1 Pump. Prior to assembling the pump to the engine, operate the pump using a dynamometer or calibrated electric motor at the designated operating speed and at a static suction lift equivalent to 10 feet. Test the pump for the requirements specified in 3.13 in accordance with the rating standards and test code standards for centrifugal pumps of the Hydraulic Institute. Data relative to head capacity shall be taken at a minimum of six points and shall include the rating points, wide-open discharge, and shutoff. Delivery of less than the capacity specified in 3.13, a peak efficiency of less than 63 percent, an NPSH greater than 8 feet, or a horsepower requirement greater than the continuous duty rating of the engine shall constitute failure of this test.

4.5.2.9.2 Pumping assembly (operational). Assemble the pump to the engine and operate the pumping assembly at the designated operating speed and at a static suction lift equivalent to 10 feet. Test the pumping assembly for the requirements specified in 3.13 in accordance with the rating standards and test code standards for centrifugal pumps of the Hydraulic Institute. Data relative to head and capacity of the pump and speed (rpm) of the engine shall be measured and recorded at a minimum of six points on the pump performance curve. The measured points shall include the rating points (see 3.13), wide-open discharge, and shutoff. The pump assembly shall run for a minimum of 5 minutes at each measured point. Delivery of less than the capacities specified in 3.13 or failure of the engine to meet any of the operational requirements or conditions specified in 3.15.1 shall constitute failure of this test.

4.5.2.9.3 Priming. Fill the pump case with either liquid petroleum fuel or a suitable solvent having a specific gravity of 0.75 and conduct the priming test with a static suction lift equivalent to 10 feet of water at standard atmospheric conditions. This test may be performed lifting either water, gasoline of 0.75 specific gravity, or a solvent of 0.75 specific gravity. Inability of the pump to prime and deliver full capacity in 2 minutes shall constitute failure of this test.

4.5.2.10 Hydrostatic pressure. Apply an internal hydrostatic pressure of not less than 200 psi to the completely assembled pumping assembly for a period of not less than 5 minutes. This test shall be performed with the 4-inch valves completely open and the quick-disconnect fitting covers in place. Nonconformance to 3.14.1 shall constitute failure of this test.

4.5.2.11 Impeller balance. During the assembly of the pump, the impeller shall be tested to determine the maximum out-of-balance condition at 2400 rpm. Nonconformance to 3.14.3 shall constitute failure of this test.

4.5.2.12 Engine performance. Prior to installation of the engine to the pump and examination and tests of the first article, the engine shall be tested in accordance with MIL-STD-1400, test method series 3000. Upon completion of this test, the engine shall be sealed in accordance with MIL-STD-1400. Nonconformance to the requirements specified herein shall constitute failure of this test.

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4.5.2.13 Fuel tank. Apply an internal air pressure of 5 psig to the fuel tank with all attachments and immerse in water so that the tank top is not less than 12 inches under the water level. Any evidence of leakage from the tank shall constitute failure of this test.

4.5.2.14 Mobility. The first article pumping assembly shall be towed over a smooth, hard-surfaced road at an average speed of 20 mph for a total of 600 miles, over a gravel road at an average speed of 10 mph for 100 miles, cross country at an average speed of 8 mph for a total of 100 miles. After each of the above segments of the test, the pumping assembly shall be operated at rated capacity for 1 hour. During the test, all the tools shall be in place in the toolbox. Evidence of bottoming, weaving, or side-swaying, inability of the trailer to follow the towing vehicle, interference between the trailer and towing vehicle, other hazardous towing characteristics, malfunction, permanent deformation, or damage to the pumping assembly tools or components, or failure of the pumping assembly to operate at rated capacity shall constitute failure of this test.

4.5.2.15 Slope. The pumping assembly shall be towed on a side slope of 20 percent. Repeat the procedure with the pumping assembly traveling in the opposite direction. Tow the pumping assembly up and down a ramp of 30 percent slope. Nonconformance to 3.19 shall constitute failure of this test.

4.5.2.16 Fording. The pumping assembly shall be towed through water at a depth of 30 inches and examined. Nonconformance to 3.20 shall constitute failure of this test.

4.5.2.17 Electromagnetic interference. The first article shall be tested to determine conformance with 3.21. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.2.18 Sling provisions.

4.5.2.18.1 Test procedure. Restrain the pumping assembly by anchoring the main frame and subject the lifting bail to lifting force of 2.5 times the weight of the pumping assembly. An acceptable alternate method of test shall be to lift the pumping assembly clear of the ground by the lifting bail and add weights to the pumping assembly until the force on the lifting bail is 2.5 times the weight of the pumping assembly. Hold the pumping assembly under either of the above loading conditions for not less than 90 seconds.

4.5.2.18.2 Failure criteria. Weld failure, permanent deformation of the pumping assembly or the lifting bail, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.19 Tiedown provisions. With the pumping assembly anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions of the tiedowns. Weld

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failure, permanent deformation of the pumping assembly or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.20 Railroad car impact. Anchor the unpackaged pumping assembly on a domestic service railway freight car (blocking is permitted). Shunt the car, with the pumping assembly tied down, at speeds of not less than 4, 6, and 8 mph, respectively, against 2 stationary flatcars having a total gross weight of at least 240,000 pounds with their airbrakes set in the emergency application position and draft gear slack removed. The pumping assembly shall face in the direction of the 2 cars. Repeat the procedure at not less than 8 mph with the rear of the pumping assembly in the direction of the 2 cars. The pumping assembly may be reversed by either reversing the direction of the flatcar (preferred) or by reloading the pumping assembly in the proper direction on the freight car. Once the test has begun, there shall be no readjustment of load nor any reconditioning of the blocks, choking material, or tiedowns, except as necessary to reverse the direction of the pumping assembly. After each impact, the pumping assembly shall be inspected. After completion of the railroad car impact test, an operational checkout of the pumping assembly shall be conducted. Nonconformance to 3.26.1 shall constitute failure of this test.

4.6 Inspection comparison. The Government may select pumping assemblies at any time during the contract production period and subject these pumping assemblies to the examination and tests specified in 4.3 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on pumping assemblies selected at random from those which have been accepted by the Government and will not include the previously inspected first article pumping assemblies. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison pumping assembly to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison pumping assembly and shall be cause for refusal by the Government to continue acceptance of production pumping assemblies until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pumping assemblies previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all pumping assemblies accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.7 Inspection of packaging.

4.7.1 First article pack inspection.

4.7.1.1 Examination. The first article pack shall be examined for defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completely processed pumping assembly prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.~~2~~2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

131.	Materials, methods, or containers not as specified for level A. Each incorrect material, method, or container shall be considered one defect.
132.	Disassembly not as specified for level A.
133.	Surfaces requiring a contact preservative in accordance with the referenced document not coated with preservative for level A.
134.	interior surfaces of the pump and connecting piping not coated with oil as specified for level A.
135.	Excess oil not drained or draincock not left open for level A.
136.	Valves not closed or dust plugs and caps not installed as specified for level A.
137.	Engine and engine accessories not preserved in accordance with the referenced document for level A.
138.	Tires not inflated as specified for level A.
139.	Repair parts not preserved in accordance with the referenced document for level A.
140.	Tools not preserved in accordance with the referenced document for level A.
141.	Components not placed in the toolbox, not cushioned, or lid not secured as specified for level A.
142.	Other removed components that will not fit in the toolbox not secured to the pump as specified for level A.
143.	Strapping not as specified for level A.
144.	Housing not provided or not secured over the pumping assembly as specified for level A.
145.	Marking missing, illegible, incorrect, or incomplete for level A or commercial.

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5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article pumping assembly or a production pumping assembly. If the first article model pumping assembly is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the first article pumping assembly and the production pumping assembly.

5.2 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. The pintle hook shall be removed and placed in the toolbox. The reflectors and any other components requiring protection against damage, pilferage, or loss, or protruding parts that increase cubage and can be removed and reinstalled without special tools or skills, shall be removed in accordance with the contractor's recommendations. Bolts, nuts, pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss.

5.2.1.2 Unprotected surfaces. Unpainted exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with, MIL-P-116.

5.2.1.3 Pump. Interior surfaces of the pump and connecting piping shall be coated with lubricating oil conforming to MIL-L-21260, grade 30. The pump shall be actuated during application of the oil to insure coating of all interior surfaces. After interior preservation is completed, the excess oil shall be drained. The draincock shall be left open. All other valves shall be closed and the dust caps and plugs installed. Exterior surfaces of the pump requiring application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.4 Engine. The engine and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, as specified for type II, method 1.

5.2.1.5 Tires. Tires shall be inflated to 10 pounds above pressure value recommended for maximum load.

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5.2.1.6 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts, and the repair parts shall be placed in the toolbox.

5.2.1.7 Operating and maintenance tools. Operating and maintenance tools shall be preserved in accordance with level A preservation and packing requirements of PPP-P-40 and placed in the toolbox.

5.2.1.8 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3, and placed in the toolbox.

5.2.1.9 Toolbox. The items contained in the toolbox shall be cushioned in a manner to prevent damage to the items. The toolbox lid shall be closed and secured.

5.2.1.10 Other components. Components removed that will not fit in the toolbox shall be secured to the pump, in a protected location that will not increase cubage or interfere with lifting or towing, with soft annealed wire or with strapping conforming to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Commercial. Commercial preservation shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.3.1 Level A. The pumping assembly shall be prepared for mobile shipment. A plywood housing consisting of two ends, two sides, and a top shall be placed over the components and extended to the trailer frame. The housing shall be constructed in accordance with the requirements of PPP-B-601, overseas type, grade B, style I, and shall be modified as required for the configuration and to permit securing to the trailer frame and shall provide cover to the components. The housing shall be secured in place by the use of lumber dunnage as required and not less than four steel straps. The straps shall conform to QQ-S-781, class 1, type I or IV, size 3/4 inch. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A. The strapping shall be applied by not less than two straps going completely around the housing from front to rear under the trailer frame and not less than two straps going around the housing from side to side under the trailer frame.

5.3.2 Commercial. Commercial packing shall be in accordance with ASTM D 3951 except the pumping assembly shall be prepared for mobile shipment.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on the shipping container.

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6. NOTES

6.1 Intended use. The type I pumping assembly covered by this specification is a component of the fuel system, supply point: portable, 60,000 and 120,000 gallon capacity, and is intended to be used for transfer and dispensing of liquid fuel. The type II pumping assembly will be used for transfer and dispensing of water.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and style required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. Time frame required for submission of first article (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. When rail impact testing is not required (see 3.26.1).
g. paint color required when other than as specified (see 3.27).
h. Time frame required for submission of first article pack (see 5.1).
i. Degree of preservation and degree of packing required (see 5.2 and 5.3).
j. When strapping shall be other than as specified (see 5.2.1.10 and 5.3.1).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the pumping assemblies in accordance with this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pumping assembly.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of pumping assemblies.

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6.7 Noise limits. When the noise level of the pumping assembly exceeds 85 dB(A), and the requirements of 3.10.1 are met, the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS paragraph) shall include the requirement for hearing protection, the noise level of the pumping assembly, the distance from the pumping assembly at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

6.8 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.9 Definitions.

6.9.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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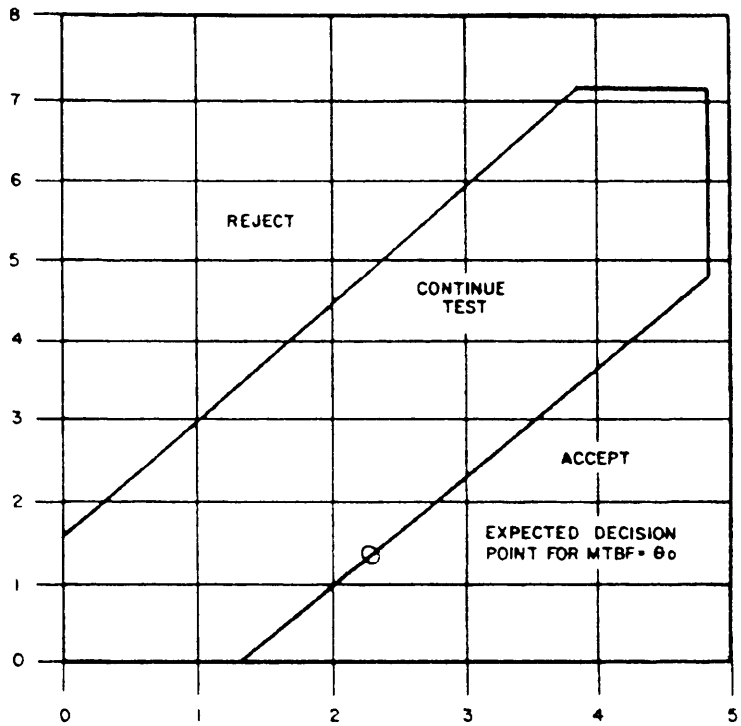
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Project 4320-A231

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DECISION RISKS 20%
DISCRIMINATION RATIO 2.0:1



NO OF FAILURES	TOTAL TEST TIME * REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3 48
4	1.73	4.17
5	2 43	4 87
6	3 12	4 87
7	3 81	4 87
8	4 87	N/A

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 5. Accept-reject criteria.

X-3661B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-52144E(ME)	Pumping Assembly, Liquid Bulk Transfer, DED, 350 GPM Capacity, 275 Total Head, Wheel Mounted

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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INCH-POUND

MIL-P-52144E(ME)
AMENDMENT 3
31 October 1994
SUPERSEDING
AMENDMENT 2
28 March 1988

MILITARY SPECIFICATION

PUMPING ASSEMBLY, LIQUID BULK TRANSFER, DIESEL-ENGINE-DRIVEN,

350 GPM CAPACITY, 275 TOTAL HEAD, WHEEL MOUNTED

This amendment forms a part of MIL-P-52144E(ME), dated 4 June 1985, and is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2, delete in its entirety and replace with the following:

"1.2 Classification. Pumping assemblies shall be of the following types and styles, as specified (see 6.2, and 6.10):

Type I - Pumping Assembly, POL.

Style A - Without regulator (TA13229E8400).
Style B - With Regulator (TA13229E8402).

Type II - Pumping Assembly, Water (TA13229E8401)."

* 2.1.1, Under SPECIFICATIONS, FEDERAL, delete:

"QQ-S-781 - Strapping, Steel, and Seals."

PAGE 2

* 2.1.1, Under SPECIFICATIONS, MILITARY, delete the following "MIL-G-1086" and "MIL-I-20241" in their entirety.

* Under SPECIFICATIONS, MILITARY, add the following:

"MIL-R-6855 - Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for."
"MIL-I-24391 - Insulation Tape, Electrical, Plastic, Pressure-Sensitive."

AMSC N/A FSC 4320
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PAGE 3

* 2.1.2, under DRAWINGS, ME, delete in their entirety and replace with the following:

"TA13229E8400 - Pumping Assembly, POL, 4 Inch, Diesel-Engine-Driven, 350 GPM Capacity, 275 Ft. Total Head, Wheel Mounted, Noise Enclosed.
TA13229E8402 - Pumping Assembly, POL, 4 Inch, Diesel-Engine-Driven, 350 GPM Capacity, 275 Ft. Total Head, Wheel Mounted, Noise Enclosed with Regulator.
TA13229E8401 - Pumping Assembly, Water, 4 Inch, Diesel-Engine-Driven, 350 GPM Capacity, 275 Ft. Total Head, Wheel Mounted, Noise Enclosed."

2.2, Under AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM), add the following:

"D 3953 - Specification for Strapping, Flat Steel and Seals."

PAGE 4

* 3.1, lines 4 and 5, delete "TA13220E1070, TA13226E2289, or TA13225E9200" and replace with "TA13229E8400, TA13226E8402, or TA13229E8401".

PAGE 5

* 3.4, delete paragraph in its entirety and replace with the following:

"3.4 Gaskets. All cork and rubber composition gaskets shall conform to MIL-C-6183, type optional, class I, grade B, except that the gasket between the intermediate and pump volute and the gasket on which the discharge check valve seats, shall be synthetic rubber conforming to MIL-R-6855, class I. Synthetic rubber gasket material shall meet all the requirements of MIL-R-6855, class I."

* 3.5, delete the first sentence and replace with "Unless otherwise specified, all threaded fasteners shall be steel and zinc coated in accordance with MIL-STD-171, finish 1.9.2.3."

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3.27, add the following new sentence at the end of the paragraph: "Top coat in accordance with MIL-C-46168 shall not be applied to engine manifolds, exhaust pipes, mufflers and other parts subject to high temperatures."

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4.4.1, delete in its entirety and replace with the following:

"4.4.1 Sampling. sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found."

* 4.4.2, delete the last sentence and replace with the following: "Presence of one or more defects shall be cause for rejection."

* 4.4.3.2, delete the last sentence and replace with the following: "Failure of any test shall be cause for rejection."

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4.5.2.2, table I, line 1, column 2 delete "X" and substitute "-" for individual pump testing.

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* 4.7.2.2, delete in its entirety and replace with the following:

"4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found."

* 4.7.2.3, delete the last sentence and replace with the following: "Presence of one or more defects shall be cause for rejection."

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5.2, delete in its entirety and replace with the following:

"5.2 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2). Preservatives applied to the interior surfaces of the pump, connecting components and any other surface that normally contacts the liquid being transferred, shall be harmless should their residual material, after evaporation, be inadvertently ingested by personnel."

5.2.1, delete in its entirety and replace with the following:

"5.2.1 Level A. Preservatives identified herein by "P" numbers shall conform to the corresponding symbols and applicable specifications in MIL-P-116."

5.2.1.3, delete the first sentence in its entirety and replace with the following: "The pump shall be drained and dried. Interior surfaces of the pump and connecting piping shall then be coated with type P-14 preservative."

Line 5, delete "left open" and substitute "closed".

5.2.1.4, add the following sentences at the end of the paragraph: "Engines with liquid cooling systems shall have the cooling system preserved by the water and antifreeze procedure as specified therein. See 5.2.1.11 for alternate fuel tank interior treatment."

* 5.2.1.10, delete in its entirety and replace with:

"5.2.1.10 Other components. Components removed that will not fit in the toolbox shall be secured to the pump in a protected location that will not increase cubage or interfere with lifting or towing of the pump. Components shall be secured using soft annealed wire or strapping conforming to ASTM D3953, type 1 or type 2, size as applicable. Finish of strapping shall be zinc coated, grade optional."

After paragraph 5.2.1.10, add the following new paragraph:

"5.2.1.11 Fuel tank. The interior of fuel tanks shall be treated with rust-preventing fog oil conforming to MIL-L-21260, grade 30, prior to the airtight sealing of all fuel tank openings in preparation for shipment or storage."

* 5.3.1, delete the fifth, sixth and seventh sentences and replace with:

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"The straps shall conform to ASTM D3953, Type 1 or 2, size as applicable. Finish of strapping shall be zinc coated, grade optional."

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* 6.2.j, delete in its entirety.

PAGE 23

Add the following new paragraph:

* "6.10 Superseding data. This specification covers two types and two styles of pumping assemblies which have been updated to include noise enclosures. The old pumping assemblies will only be used as follows:

"Type I - Pumping Assembly, POL

Style A (TA13220E1070)-
Inactive for new design, use for procurement of repair parts only.

Style B (TA13226E2289)-
Inactive for new design, use for procurement of repair parts only.

Type II - Pumping Assembly, Water

(TA13225E9200)-
Inactive for new design, use for procurement of repair parts only."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Army - ME

Review activity:
DLA - CS

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